

**CSE 360: Computer Interfacing**  
**Department of Computer Science and Engineering**  
**Brac University**  
**Semester: Spring 2026**  
**Assignment 1**  
**[Mark 5]**

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A **digital weather station** is designed using an **STM32 microcontroller**. It collects data from a **temperature sensor** which is connected with **pin 4** and **6** of **port B**, shows the reading on a **digital display** which is connected with **pin 4, 8,9** and **10** of **port E**, and sends it to a **mobile app via a wireless module** which is connected with **pin 2** and **3** of **port F**.

The designer chose three different communication methods:

- One uses **two lines** (data and clock) to connect the sensor,
- One uses **a master-slave setup with separate data-in, data-out, clock, and chip-select lines** for the display, and
- One uses **a simple serial link with start and stop bits** for wireless data transmission.

**Questions:**

1. Identify the **communication protocol** used in each device.
2. State whether each is **serial or parallel** and **synchronous or asynchronous**.
3. Justify briefly **why each method suits the specific device**.
4. Write down **all the necessary commands** for initial configuration of the system.